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530:0100

GENERAL

Materials, components, workmanship, inspection and testing shall be in accordance with this specification, Local Authority Regulations and B.S. 6700:1987 and standards & publications referred to therein, unless otherwise stated. All pipework, fittings and valves shall be approved to British Standard kite mark or be produced under ISO 9000.

530:0200

TUBES AND FITTINGS

530:0210 Domestic hot water service pipework shall be half hard, light gauge PPR to BS 4991/6920 or DIN 8077/8078/16962 Part 1 to 12.

530:0220 Domestic cold water service pipework above ground shall be half hard, light gauge PPR to BS 4991/6920 or DIN 8077/8078/16962 Part 1 to 12 to meet the pressures required. Minimum pressure rating for pipe to be 20 bar. Domestic cold water service pipework below ground shall be shall be uPVC to B.S.3505 Class "E".

530:0230 Water tube shall be mild steel to either B.S. 1387, heavy grade (red band) seamless black vanished, or B.S. 3601 reference S-320 black vanished, seamless.

530:0240 PR fittings shall be socket or butt fusion type suitable for heat fusion. All socket fusion fittings shall exhibit an interference fit prior to joining with dimensions and tolerances in accordance with ISO 7279 and DIN 16962 Parts 1 to 12 and related ISO Recommendation and Draft Resolutions. All butt fusion fittings shall be manufacture in accordance with DIN 16962 Parts 1 to 12 and related ISO Recommendation and Draft Resolutions. All fittings shall be related to 20bar.

530:0250 uPVC tube fittings shall be to B.S. 4346; Part 1 solvent welded type complete with stub flange or screwed adapter change fittings when connecting to other materials.

530:0260 50mm N.B. mild steel fittings shall be malleable cast iron banded to B.S. 143 and 1256, black vanished, with B.S 21 tapered thread minimum pressure rating 20 bar. Above 50mm N.B. mild steel fittings shall be butt- welded type to B.S 1965: Part 1 heavy grade, with chamfered ends minimum pressure rating 20 bar.

530:0270 Flanges shall be to B.S 4504: Part 1, black vanished.

530:0280 The pipe sizes referred to on the drawing are nominal. The actual pipe sizes shall be as below:

SHARJAH ASSET MANAGEMENT HOLDING
SOUQ AL HARAJ (SQAH) PROJECTS
PACKAGE 2
VOLUME II - SPECIFICATIONS

Black iron nominal bore (mm)	Black iron internal bore (mm)	Copper nominal bore (mm)	Copper internal bore (mm)	PP-R nominal bore (mm)	PP-R internal bore (mm)	uPVC nominal bore (mm)	uPVC internal bore (mm)
15	14.9	15	13.6	16	11.6	15	17.2
20	20.4	22	20.22	20	14.4	20	21.8
25	25.7	28	26.22	25	18	25	28.2
32	34.4	35	32.63	32	23	32	35.9
40	40.3	42	39.63	40	28.8	40	40.9
50	51.3	54	51.63	50	36.2	50	51.4
65	67	67	64.27	63	45.6	65	64.2
80	79.1	75	73.22	75	54.2	80	75.9
100	103.5	110	105.12	90	65	100	97.7
125	128.85	133	130.38	110	79.6	125	120
150	154.3	160	155.38	125	90.8	150	144
175	183			160	116.2	175	165.9
200	207.3					200	190.9
250	260.4					250	238

530:0300 ISOLATING VALVES

530:0310 For sizes up to and including 50mm shall be bronze (to BS1400 LG2) gate valve to BS5154 PN20 Series B.

Valves shall be complete with position indicator and locking device, asbestos free gland packing, with non- rising stem, threaded bonnet, one piece wedge and BS21 taper thread (ISO R7). Valves shall be Kite marked.

530:0320 For sizes 65mm and above shall be cast iron (to BS1452 Gr 220) gate valve to BS5150 PN16. Valves shall be complete with locking device and open shut indicator, rising stem, outside screw, zinc free bronze trim, wedge facing ring and body seat ring shall be of gunmetal, flanged to BS4504 PN16. Valves shall be internally and externally epoxy coated and Kite marked.

530:0330 Potable water isolating valves sizes 65mm and above shall be cast iron (to BS1452 Gr220) gate valve to BS5163 PN16. Valves shall be fitted with indicator and supplied with two bosses drilled and plugged, non- rising stem, inside screw, wedge facing ring and body seat ring shall be of gunmetal, flanged to BS4504 PN16. Valves shall be internally and externally epoxy coated and Kite marked.

530:0400 REGULATING VALVES

530:0410 Regulating valves shall be of the double regulating “Y” type, lock shield pattern with none rising stems.

530:0420 50mm N.B. and below regulating valves shall be to B.S. 5154 and shall have B.S 1400-LG2 gunmetal body.

Glands shall be lubricated packed type, capable of re-packing under pressure. Disks shall be characterized to give a linear flow regulating performance. The bonnet shall be of the screwed type. Pressure rating shall be PN20. Connections shall be screwed to B.S 21 tapered thread.

530:0430 65mm N.B. and above regulating valves shall be to B.S. 5152 and shall have B.S 1452 Gr 220 cast iron body.

Glands shall be lubricated packed type, capable of re-packing under pressure. Disks shall be characterized to give a linear flow regulating performance. Pressure rating shall be PN20. Connections shall be flanged to B.S 4505 table 16/11. The valve shall be painted to protect it from corrosion.

530:0440 Fixed orifice valves shall consist of a matched assembly of a regulating valve, orifice plate and test plugs.

530:0450 The Contractor shall hand over to the Engineer two (2) regulating valve keys, per fifty (50) installed regulating valves.

530:0500 CHECK VALVES

530:0510 Check valves shall be of the swing pattern, low pressure loss, suitable for horizontal and vertical installations.

530:0520 25mm N.B. and below check valves shall have B.S 1400-LG2 gunmetal body. The disk and pin shall be manufactured from brass to B.S. 2874 CZ121. Pressure rating shall be PN20. Connections shall be screwed to B.S 21 tapered thread.

530:0530 32mm N.B. to 50mm N.B. check valves shall have B.S 1400-LG2 gunmetal body. The disk and pin shall be manufactured from gunmetal to B.S. 1400LG2. Pressure rating shall be PN20. Connections shall be screwed to B.S 21 tapered thread.

530:0540 65mm N.B. and above check valves shall have B.S 1452 Gr 220 cast iron body and disk. The pin shall be manufactured from H.T. Brass to B.S. 2874 CZ114. The disk shall be solid type with gunmetal facing ring. Pressure rating shall be PN20. Connections shall be flanged to B.S 4505 table 16/11. The valve shall be painted to protect it from corrosion.

530:0600 BALL VALVES

530:0610 Ball valves shall have a gunmetal body, with stainless steel ball and screwed to B.S 21 tapered thread.

530:0700 FLOAT VALVES

530:0710 Float valves shall be installed as indicated in the drawings to provide consistent level control in reserve supply water storage tanks. The valve shall meet the requirements of the Local Authorities and shall prevent back siphonage. Inlet flow shall be co-axial with the piston movement, to ensure high discharge capacity and smooth, quiet operation. The valve shall have 'O' ring piston seals, resilient seated disk, and Dezincification resistant bronze body and disk components. All floats shall be manufactured in copper to BS2870 and fabricated using approved solder and shall be hydraulically tested to ensure buoyancy and constructed in accordance with BS1968.

530:0800 DRAIN COCKS

530:0810 Drain Cocks shall be lock shield pattern to BS 2879:1980, Type 2. The drain cock body and all parts in contact with water are to be made from materials resistant to dezincification

530:0900 STRAINERS

530:0910 Strainers shall be "Y" type, complete with either stainless steel or brass screens.

530:0920 50mm N.B. and below strainers shall have B.S 1400-LG2 gunmetal body. Pressure rating shall be PN20.

Connections shall be screwed to B.S 21 tapered thread.

530:0930 65mm N.B. and above strainers shall have B.S 1452 Gr 220 cast iron body. Pressure rating shall be PN20.

Connections shall be flanged to B.S 4505 table 16/11. The valve shall be painted to protect it from corrosion.

530:1000 GAUGES

530:1010 Temperature gauges shall be 100mm diameter, lacquered brass body, scale selected so that the normal operating point is approximately half scale. Gauge shall be complete with pipe insertion pocket. Temperature gauges will be rated in 0C.

Gauges shall be fixed so that the tails are truly subjected to the temperature of the water to be measured. Right angle or obtuse angle thermometers shall be provided for fittings to vertical pipes or vertical faces for cylinders etc. Thermometers shall be dial type.

530:1020 Pressure gauges shall be 100mm diameter, lacquered brass body, scale selected so

that the normal operating point is approximately half scale. Gauge shall be complete with isolation cock. Gauges shall be calibrated in bars and kPa.

530:1030 Thermometers shall be mercury in glass with protective shroud, scale selected so that the normal operating point is approximately half scale.

Thermometers shall have 15mm BSP threads and perforated steel tail measuring 75mm from the underside of the hexagon. The scale shall be boldly marked on the glass and shall be graduated from 0 c to 40 c and shall be accurate to plus or minus 2% over the total range. A well shall be supplied for each thermometer for insertion into the waterway and shall be provided with sufficient approved heat conducting medium. The well shall be mild steel but shall be brass for copper installations.

530:1100 EXPANSION TANK

The Expansion Tank shall be sized as required for the system volume and shall conform to BS 7074. Tank shall be constructed and stamped for pressure required for the working pressure in the particular system.

– Diaphragm Type Pre-Pressurized Tanks.

Tank shall be provided with an integral butyl rubber diaphragm which shall contain a sealed-in air cushion pre- charged to the minimum system pressure. Tank shall have an air charging valve and shall be designed to a maximum operating temperature required in the particular system. All components in contact with domestic water shall be suitable for use with products for human consumption.

530:1200 IIAHUAL DOSING POT

The unit shall be a steel one, the design pressure in accordance to the system pressure. Feeder shall be supplied complete with fill and drain valves, filling funnel, and air cock.

530:1300 SUPPORTS

530:1310 FIXING OF SUPPORTS

The Contractor shall securely fix all supports, brackets, hangers etc. with approved fixings suitable for the building structure.

530:1315 SUPPORTS

All supports shall be of proprietary type suitable for the pipe material and location. Pipe supports exposed to view shall be of the easy clean type. Pipe supports in concealed locations shall be of the split ring type.

530:1320	CLEARANCE
	The Contractor shall ensure that all pipes have a minimum clearance of a half pipe diameter from the finished face of the structure.
530:1325	PIPE THRUST
	The Contractor shall provide full details of the method of locating and anchoring the pipework to the Engineer for approval before fixing is commenced.
530:1330	PIPE MOVEMENT
	The Contractor shall support all pipework adequately and in such a manner as to disallow any movement of the tubes other than longitudinal movement due to expansion and contraction, by allowing a marginal clearance between the pipes and the supports.
530:1335	LOADING
	The Contractor shall ensure that the load on any hanger or support in no case exceeds the manufacturer's recommended maximum load with respect to the fixing component.
530:1340	FINISH
	The Contractor shall grind all cut ends of any pipework support to give a neat round finish to the satisfaction of the Engineer.
530:1345	PAINTING OF SUPPORTS
	The Contractor shall paint all ferrous components of supports, hangers, brackets, pipe clips, etc., on completion of the installation with one coat of anti-corrosive primer and one top finishing coat. Where pipe supports/ secondary steelwork are located in an external ambient environment (including internal car parks or naturally ventilated plant rooms) all supports shall be galvanised after manufacture.
530:1350	PIPE RINGS
	Copper pipework shall be supported using standard brass or gunmetal, split pipe ring assemblies, with the two halves of the pipe ring secured by bolts and nuts or screws. Plastic pipework shall be supported using either proprietary plastic clips or galvanised pipe rings with rubber inserts. Mild steel pipework shall be supported using rings in accordance with B.S 3974:1974.
530:1355	WALL BRACKET

For small diameter pipework in exposed locations, the Contractor shall use pipe clips with integral wall back plates.

530:1360 MAXIMUM DISTANCE BETWEEN PIPE SUPPORTS

Scheduled below are the maximum spacing for pipe supports.

Pipe N.B. (mm)	Vertical Installation.			Horizontal Installation.		
Material	Copper	Steel	Plastic	Copper	Steel	Plastic
15	1.8	2.4	1.0	1.2	1.8	0.50
20	1.8	3.0	1.0	1.2	2.4	0.50
25	2.4	3.0	1.5	1.7	2.4	0.75
32	2.4	3.0	1.5	1.7	2.4	0.75
40	2.6	3.7	1.5	2.0	2.4	0.75
50	2.6	3.7	2.0	2.0	2.4	1.00
65	2.6	4.6	2.0	2.0	3.0	1.00
80	3.0	4.6	2.0	2.4	3.0	1.00
100	3.3	4.6	2.0	2.7	3.0	1.00
150	-	5.5	2.0	-	4.5	1.00
200	-	8.5	-	-	6.0	-
250	-	9.0	-	-	6.5	-
300	-	10.0	-	-	7.0	-

530:1400 WORKMANSHIP

530:1410 OPEN ENDS

The Contractor shall plug open ends of pipes to prevent entry of foreign matter into the system during construction.

530:1411 CUT ENDS

The Contractor shall cut ends of pipes, gutters, fittings etc., clean and square using equipment appropriate to the material.

530:1412 BENDS & BRANCHES

The Contractor shall ensure that all changes in direction are made using proprietary fittings and are installed such that they curve in the direction of flow.

530:1413 PIPE BENDING (COPPER)

The Contractor may bend using an approved device, pipes up to 22mm sizes where the bends are concealed from view.

- 530:1414 **GRADING OF PIPEWORK**
- The Contractor shall grade all horizontal pipework to allow for venting and draining.
- 530:1415 **SPACING OF PIPEWORK**
- The Contractor shall allow the following minimum clearances between lines of adjacent pipework and between pipework and the building structure, unless otherwise instructed.
- 530:1416 **DISSIMILAR MATERIALS**
- The Contractor shall ensure that all joint in pipes of dissimilar materials are as recommended by the manufacturer. The Contractor shall ensure that materials liable to cause corrode touch each other.
- 530:1417 **JOINTING MATERIAL**
- The Contractor shall ensure that jointing materials do not project into the bore of the pipes, fittings or appliances.
- 530:1418 **CLEANLINESS**
- The Contractor shall ensure that he cleans and dries all jointing faces before preparation and application of the joint.
- 530:1419 **INJURIOUS COMPOUND**
- The Contractor shall not use compounds containing red lead or any other injurious or tainting elements.
- 530:1420 **HYGIENE**
- The Contractor shall ensure that all pipes are kept clean to the degree that general site conditions allow. Coat all rubber joint rings with bactericidal compound before use.
- 530:1421 **THREADS**
- The Contractor shall protect all exposed threads of ferrous pipes above ground with an approved paint.
- 530:1422 **FLANGED CONNECTIONS**
- The Contractor shall install flange connections every 14m along pipelines to act as dismantling points, at all valves or where pipework is connected to an appliance, and

where indicated.

530:1423 SCREWED FLANGES

The Contractor shall ensure that screwed flanges are screwed to the pipe tight home to finish flush with the flange face, and remove all surplus material projecting beyond the flange face, with a fine file and dress the face smooth.

530:1424 BRAZED FLANGES

Braze gunmetal flanges to copper tubes with good quality silver brazing alloy in accordance with B.S. 1845, Type 3.

530:1425 FLANGE BOLTS

Bolted flanged joints shall be with hexagon head 8.8.150 (minimum) metric coarse thread, high tensile steel bolts and nuts to B.S. 6104, complete with one flat steel washer on either side of the paired flanges. The Contractor is to ensure that the projections of the bolts are no more than four threads and not less than three threads beyond the nut when tightening up.

530:1426 FLANGE JOINTING RINGS AND GASKETS

Shall use flat ring gaskets extending to the inside of the bolt circles for flange joints with flanges complying with B.S. 4504.

530:1427 BLANK FLANGES

Shall be blank flanges of true circles with smooth edges of the same diameter and drilled to match the flange to which it is fitted. The surfaces shall be machined flat and free from all irregularities.

530:1428 EMBEDDED PIPEWORK

Where it is necessary to embed pipework, it shall be wrapped in flexible inert foam. The Contractor shall ensure that no part of the embedded pipework is allowed to come in contact with the mortar.

530:1429 SLEEVES

The Contractor shall install sleeves around pipes when they pass through walls and the structure. The pipe sleeves shall project 3mm beyond the finished face of the building structure, and shall have a diameter not less than 10mm larger than the outside diameter of the passing through pipe including insulation. The Contractor shall pack the free space between the pipe and the sleeve with a suitable non-

combustible material to provide an effective seal.

530:1430 AIR VENTS

The Contractor shall install automatic air vents at the high points in the system. Automatic air vents in main risers shall be 15mm N.B. complete with isolation valve and manual bypass valve to allow for quick venting.

530:1431 TUNDISHES

The Contractor shall install tundishes where drip pipes from pumps, discharge to waste or collect into common drains, unless the discharge is direct to a floor gully.

530:1432 FLEXIBLE PIPEWORK CONNECTIONS

The Contractor shall install flexible connectors between moving plant and the pipework.

530:1433 SAFETY VALVES

The Contractor shall select safety valves so that the discharge setting is in compliance with the pressure calculated in BS 759. Safety valves discharges shall be copper pipe of equal bore to the inlet connection and routed to a safe visible location.

530:1434 WELDS

The Contractor shall demonstrate to the Engineer the capability of his welders by producing representative test welds. If during the installation the Engineer considers it necessary the Contractor shall carry out non- destructive tests on welds as directed by the Engineer.

530:1500 TESTING

530:1510 NOTIFICATION OF TESTS

The Contractor shall inform the Engineer in writing at least three (3) working days before any test.

530:1511 SUPERVISION AND LABOUR

The Contractor shall provide any personnel required for carrying out and supervise the tests, as well as all equipment, appliances and materials for interim testing throughout the progress of the work and for final and performance tests. Fix tee-pieces and plugs when and where required and seal off and remove as directed by the Engineer.

530:1512 OBSTRUCTIONS

The Contractor shall ensure that all pipework is properly secured and clear of obstructing debris and superfluous matter and that all work which is to be concealed is free from defects before it is finally enclosed.

530:1513 TEST INSTRUMENTS

The Contractor shall supply all necessary apparatus and measuring instruments etc. for the specified tests, and ensure that the equipment is in good working order and calibrated to the satisfaction of the Engineer.

530:1514 DEFECTS

The Contractor shall rectify any defects of workmanship, materials and equipment that have become apparent during the tests, and repeat any relevant part of the tests, as necessary, to the satisfaction of the Engineer.

530:1515 TEST CERTIFICATES

The Contractor shall maintain test certificates. He shall ensure that:

1. At the satisfactory completion of any of the specified tests the results are recorded on an agreed test certificate.
2. Arrange for working copies of all signed certificates to be forwarded to the Engineer as soon as practicable after the completion of the tests.
3. Bind the completed test certificates into the operation and maintenance manuals.

530:1516 DEFINITIONS

1. Off-Site Tests: tests carried out on items of plant and equipment at manufacturers' works or elsewhere to ensure compliance with the requirements of the specification and or relevant British Standard or Codes of Practice.
2. Site Tests: tests on static plant and systems (e.g. inspection and testing of welds, hydraulic testing of pipework etc.), to ensure correct and safe installation and operation.

530:1517 The Contractor shall submit to the Engineer, Test Certificates for all tests carried out at the manufacturers' works or elsewhere, as appropriate tests on items of equipment to demonstrate compliance.

Note that in certain cases Type Test Certificates will be accepted as follows:

1. Pumps: Type Test Certificate for head, discharge, speed and power input in accordance with B.S. 599.
2. Electric Motors: Type Test Certificates in accordance with B.S. 5000, and Routine Test Certificates in accordance with B.S. 5000 for motors of 40 kW output and above.
3. Starters and Control Gear: Type Test Certificates in accordance with B.S. 587 and Routine Test Certificates for high voltage in accordance with B.S. 587 for control panels as a whole.

530:1518 ISOLATING VALVES

If the section of pipework to be tested terminates at an isolating valve, the valve shall not be used to retain the water. The Contractor shall fit a blanking flange or a plug to the valve temporarily.

530:1519 HOTWATER HEATERS

Hot water heaters shall be hydraulically pressure tested at one and a half times the normal working head with all mountings (except the relief valve) and connections complete. The test pressure to be maintained for 30 minutes after making good any leaks. Test relief valves separately. The Contractor must check the safe test pressure of the above equipment prior to carrying out any test.

530:1520 INITIAL FILLING

The Contractor shall slowly fill with clean water to exclude all air. Apply a test pressure measured at the inlet of one and a half times the normal working head to be maintained for 30 minutes after making good any leaks. Ensure that plant and equipment that is unable to withstand the test pressure is disconnected from the pipeline during the test.

530:1521 TEST PRESSURES

The Contractor shall hydraulically test the pipework to 700kPa or 1.5 times the working pressure whichever is greater for a period of at least 2 hours.

530:1522 CIRCULATION SYSTEMS

The Contractor shall employ a specialist sub-contractor to test and commission all circulation systems in accordance with the C.I.B.S.E commissioning codes. Refer to specification 584.

530:1600 STERILISATION

530:1610 The Contractor shall sterilise all domestic water service pipes, as well as tanks and water heaters, before the system is brought into use. The sterilisation shall be carried out by a specialist sub-contractor, who shall:

1. Clean and flush out to remove dirt and debris. No water is to be used for domestic purposes neither during this operation nor until sterilisation is complete.
2. Sterilise new mains and service pipes first, followed by tanks and distribution pipes in that order.
3. Prepare the necessary quantity of sterilising chemical to give chlorine concentration of 50mg/l. Follow the manufacturer's instructions.

Note: Using hypochlorite having 35% available chlorine, some 145 grams/100.1 of water to be sterilized will be required.

- 4 Empty the section of pipework, or tank to be sterilised.
- 5 Inject the prepared solution at the lowest level and fill the pipework from that point with clean water.
- 6 Allow a contact time of 2 hours when the section is full.
- 7 Take a sample at the end of the time and check for residual chlorine with a comparator to the satisfaction of the Engineer. If less than 0.2mg/l residual is indicated, repeat the process until the residual is obtained.
- 8 Thoroughly flush out to remove all traces of heavily chlorinated water.

530:1700 LABELLING

530:1710 The Contractor shall identification of all items of plant and equipment and pipework, by means of either label, colour coding, P.V.C. adhesive tape, numbered ferrules, and terminal markings.

530:1720 SAMPLES

The Contractor shall provide samples of all identification labels and tapes as well as a statement regarding the proposed methods of application, prior to the commencement of the work.

530:1730

EXPLANATORY CHARTS

The Contractor shall provide an explanatory chart for the use of the Employer's Maintenance Staff, showing the colour codes used on the installations, as well as all other symbols, figures and letter codes incorporated on the identification tapes and labels.

530:1740

VALVE CHART

The Contractor shall provide a valve chart giving the description and use of each valve forming part of a pipework installation, as well as the valve

530:1750

COLOUR CODING AND LETTERING

The Contractor shall fit all pipework in plant rooms, false ceilings and service risers and ducts with identification tapes give the following information:

1. Colour coding, where applicable, in accordance with the recommendations of B.S. 1710.
2. Name of service, whether flow or return, and an arrow indicating the direction of flow, all clearly legible against the background colour and with lettering not less than 6 mm high.

530:1760

SPACING OF TAPES

The Contractor shall place the identification tapes at all pipework junctions, at both sides of valves and items of plant, at floor, ceiling and wall penetrations, and at not less than 5 m intervals on all straight pipe runs.

530:1770

VALVE LABELS

The Contractor shall fit valve labels to identify all valves and cocks by permanently fixed labels, manufactured from "Trifoliolate" or similar material, and engraved with markings and/or lettering in accordance with an agreed system of valve reference.

530:1780

MANUFACTURERS' NAME PLATES

The Contractor shall ensure that all items of plant are provided with a securely fixed manufacturers' name plate, giving the reference number, size and model number etc. of the plant, as well as the following information, as applicable.

- 1 Speed in r.p.m., horse power, full load running and starting currents and voltage in the case of electrical plant.

2. Any other details necessary to allow the item of plant to be readily identified for the purposes of future replacement and spare parts.

530:1800

AIR ELIMINATORS

Air vents shall be installed on all coils and all other high points required for efficient operation and venting of system. Air vents shall be provided at all high points in the pipework, whether indicated on the drawings or not.

Large diameter bottle type automatic air vents shall be provided at all primary venting positions, such as plant rooms and at the head of vertical risers.

Heating and cooling coils on air handling units, fan coil units and ductwork systems, in addition to all high points of pipework which require venting, shall be provided with 6 mm diameter air cocks unless some other form of venting is specified or has been indicated on the drawings.

For coils and similar cased appliances, the air cock shall be located for convenient operation external to the outer casing.

Air bottles shall be provided at all venting points. Air bottles shall be formed from pipe of equal bore to the pipe being vented and of length equal to the bore plus 150 mm. Each air bottle shall be provided with a 6 mm vent pipe welded into the top and terminating in a conveniently accessible position, approximately 1.600 m above floor level, with a 6 mm lock shield needle valve. The needle valve shall be provided with a 6 mm tailpiece with mitred end.

Each automatic air vent shall be preceded by a lock shield pattern stop valve, and the discharge from the air vent shall be 12 mm copper pipe terminating with an open discharge in a position to be agreed over a conveniently located drain, gulley or sump.

The Contractor shall be responsible for the design and positioning of all air vents. The Contractor shall submit details of sizing, selection and location to the Engineer for review prior to installation.

Automatic Vents

- a. For fire protection systems (i.e. sprinklers and wet risers) the vents shall be UL listed/FM approved and approved by the Civil Defense Authority.
- b. Vents on air separators shall be float actuated designed to purge free air from the system and provide positive shut-off at the working pressures kPa at a maximum working temperature. Vents shall be tightly sealed against loss of

system water and prevent entrance of air in negative pressure situations. Vents shall be constructed of cast iron and fitted with components of stainless steel, brass, EPDM, and silicone rubber.

Manual Vents

The vents on coils and other mechanical equipment shall be as per approved equipment manufacturer's standards. Only these may be manual type and all other vents shall be automatic.

Automatic air vents shall have gunmetal or brass bodies, non-ferrous or stainless steel floats, guides and non-corrodible valves and seats. Each automatic air vent shall be controlled by a lock shield valve and a non-return valve shall be fitted on air release pipe. Air release pipes shall be run to discharge at the nearest suitable point visible outside the building. AAV shall be 15mm dia. ball float type with 15mm copper drain pipe.

Manual air cocks, where specifically called for on the drawings shall be nickel-plated of the spout less pattern with screwed taper thread. Four loose keys shall be provided for each installation having up to ten air cocks and one loose key shall be provided for every additional ten air cocks.

It is the responsibility of the contractor to install all pipework graded to a vent point to ensure a proper hydraulic flow in the system. Any part of the system that does not vent to a natural vent point shall be replaced at no cost to the contract.

Air vents shall be installed as specified in positions agreed with the Engineer prior to the completion of the installation.